# MCSOA: A Novel High-Reliable Wireless Sensor Networks Routing Optimization Method for Power Internet of Things


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***Abstract*—In recent years, as the Power Internet of Things continues to advance and improve, the routing requires efficient and reliable optimization methods to ensure stable operation. However, sensor nodes in Wireless Sensor Networks for Power Internet of Things have limited energy and lifespan, making the search for an efficient routing optimization method a critical issue. This paper innovatively proposes a novel high-reliable Wireless Sensor Networks routing optimization method for Power Internet of Things—the Multi-Objective Chaotic Seagull Optimization Algorithm (MCSOA). MCSOA integrates chaotic optimization strategies with the Seagull Optimization Algorithm, enable to escape local optima and improving its global search capability. Additionally, it incorporates multi-objective functions to address various key factors in PSWSN routing, including minimizing transmission loss, balancing load distribution, and enhancing system reliability. To evaluate the effectiveness of the optimization method, extensive new experiments were conducted, accurately reflecting real-world power grid operation scenarios. The experimental findings indicate that MCSOA significantly outperforms traditional algorithms like Genetic Algorithm and Particle Swarm Optimization in terms of efficiency. Specifically, MCSOA cuts down energy use by a minimum of 16.1% and prolongs the lifespan of the PSWSN by at least 15.6%.**



***Keywords*—Wireless Sensor Networks, Power Internet of Things, Routing optimization, Seagull Optimization algorithm**


## I. INTRODUCTION

Wireless sensor networks are crucial in the ongoing advancement of the Power Internet of Things. In PIoT, WSNs are utilized for monitoring and managing power distribution, ensuring efficient energy consumption, and preventing system failures. Recently, wireless sensor networks in power systems (PSWSNs) have garnered increasing research interest and attention [1][2].

A key challenge in designing PSWSNs is optimizing the routing process to ensure networks' high reliability and efficiency[3]. Given that sensor nodes are typically equipped with limited power and lifespan, the routing optimization significantly impacts the overall performance of the network. So, it's essential to enhance the routing efficiency to maximize the network's reliability and lifespan. In PSWSNs, routing optimization is crucial for enhancing network performance [4][5]. But, determining the best routing paths is an NP-hard problem, which complicates achieving the optimal solution due to the high computational complexity of exhaustive search methods is impractical for real-world applications. Traditional deterministic algorithms often fail to provide effective routing optimization methods in large-scale PSWSNs. With the deepening research on swarm intelligence heuristic algorithms, various optimization algorithms have emerged, such as the Genetic Algorithm[6] and Particle Swarm Optimization[7]. However, these routing optimization algorithms, which simulate the intelligent behavior of biological populations, often fall into local optima due to premature convergence when obtaining the optimal routing scheme. This situation can increase network energy consumption and shorten network lifetime.

Therefore, this paper innovatively proposes a new high-reliability wireless sensor network routing optimization method for the Power Internet of Things, namely the Multi-Objective Chaotic Seagull Optimization Algorithm (MCSOA), which significantly extends network lifetime and reduces node energy consumption. Additionally, the introduction of chaotic optimization strategies and seagull optimization algorithms enhances the algorithm's ability to escape local optima, improving global search capabilities.

To achieve these goals, this research makes the following contributions:

- Firstly, it proposes MCSOA for solving the routing optimization problem in WSNs for the PIoT. M-

MMCSOA combines multiple objective functions to address various key factors in the routing of Power Internet of Things Wireless Sensor Networks (PSWSNs), including minimizing transmission loss, balancing load distribution, and improving system reliability, significantly reducing system energy consumption, and effectively extending the lifespan of PSWSNs.

- Secondly, MCSOA combines chaotic optimization strategies and seagull optimization algorithms, balancing balancing local and global convergence, thereby improving both accuracy and speed. The global search capability of MCSOA ensures the identification of optimal routing solutions under different conditions of Power System Wireless Sensor Networks.
- Finally, extensive simulation experiments are conducted to compare MCSOA with Genetic Algorithm and Particle Swarm Optimization methods in routing optimization of power systems. The experimental results show that MCSOA outperforms GA and PSO, demonstrating superior routing efficiency and system stability. The proposed MCSOA method is not only more powerful but also simpler than existing heuristic algorithms, effectively avoiding local optimization and producing better results.

The structure of this paper is as follows: Section 1 covers the research background. Section 2 introduces the innovatively designed PSWSNs model. Section 3 proposes the MCSOA algorithm for PSWSNs. Section 4 compares MCSOA with GA and PSO through simulations. Section 5 summarizes the work and outlines future research directions.

## II. MODEL DEFINITION

### A. Network model

This section introduces a mathematical model for routing optimization in WSNs used in the IoT for power grids. The network structure can be abstracted as a graph represented by Gr(Ve, Ed). Here, Ve denotes the set of all sensor nodes a , where Ve={v1→…→vp→…→va} and Ed represents the set of links between nodes, denoted as Ed={vp→vq}(p≠q) indicating the communication relationships between nodes[8].

### B. Energy model

The energy consumption of links in PSWSNs includes the energy used for data transmission and reception, with the total energy consumption defined as:

$$To_{EC} = Se_{EC} + Re_{EC} \tag{1}$$

In (1), where:

$Se_{EC}$ denotes the energy used for data transmission,

$Re_{EC}$ denotes the energy used for data reception.

$$Se_{EC} = \begin{cases} \alpha \cdot En_{ele} + \gamma_{apx} \cdot \alpha \cdot dist^2 & dist \le dist_0 \\ \alpha \cdot En_{ele} + \gamma_{apy} \cdot \alpha \cdot dist^4 & dist > dist_0 \end{cases} \tag{2}$$

$$Re_{EC} = \alpha \cdot En_{ele} \tag{3}$$

In (2) and (3), where:

*dist* represents the communication distance,

$\alpha$ stands for data transmission amount,

$En_{ele}$ denotes the energy required for transmission and reception circuits,

$\gamma_{apx}$ represents the power amplification factor for free-space transmission,

$\gamma_{apy}$ represents the power amplification factor used for multipath attenuation transmission.

### C. Objective Functions

To achieve the desired optimization goals, we define the following objective functions:

*a)* Minimizing Transmission Losses:

$$f_1 = \sum_{i=1}^{N} \sum_{j=1}^{N} P_{ij} R_{ij} L_{ij} \tag{4}$$

In (4), where:

$N$ is the count of nodes in the WSNs for PIoT.

$P_{ij}$ is the power transmitted between node $i$ and node $j$.

$R_{ij}$ is the transmission line resistance between nodes $i$ and node $j$.

$L_{ij}$ is the length of the transmission line between node $i$ and node $j$.

*b)* Balancing Load Distribution:

$$f_2 = \sum_{i=1}^{N} \left| P_i - \bar{P} \right| \tag{5}$$

In (5), where:

$Pi$ is the load at node i.

$\bar{P}$ is the average load of the system.

*c)* Enhancing System Reliability:

$$f_3 = -\sum_{i=1}^{N} \sum_{j=1}^{N} R_{ij} \times A_{ij} \tag{6}$$

In (6), where:

$A_{ij}$ is the availability of the transmission line between node $i$ and node $j$.

*d)* Power Flow Constraints:

$$\sum_{j=1}^{N} P_{ij} = P_i \quad \forall i = 1, 2, ..., N \tag{7}$$

This ensures that the power flow balance is maintained at each node.

*e)* Capacity Constraints:

$$P_{ij} \le C_{ij} \quad \forall i, j = 1, 2, ..., N \tag{8}$$

In (8), where:

$C_{ij}$ is the capacity of the transmission ways between node $i$ and node $j$.

*f)* Voltage Constraints:

$$V_{\min} \le V_i \le V_{\max} \quad \forall i = 1, 2, ..., N \tag{9}$$

where:

$V_i$ is the voltage at node $i$, and $V_{min}$ and $V_{max}$ are the minimum and maximum allowable voltages, respectively.

*g)* Network Topology Constraints:

$$A_{ij} \in \{0,1\} \qquad \forall i,j = 1,2,\ldots,N \tag{10}$$

In (10), where:

$A_{ij} = 1$ if there is a transmission line between node i and node j,

$A_{ij} = 0$ otherwise.

*h)* Multi-objective Optimization Fitness Function

To evaluate the performance of the proposed MCSOA algorithm, a composite fitness function is used, incorporating the weighted sum of the objective functions:

$$Fitness(x) = w_1 f_1(x) + w_2 f_2(x) + w_3 f_3(x) \tag{11}$$

In (11), where:

$w_1$, $w_2$, and $w_3$ are the weights assigned to each objective function, reflecting their relative importance in the optimization process.

In summary, the model for routing optimization in Wireless Sensor Networks for Power Internet of Things integrates multiple objectives and constraints to reflect the complex nature of PSWSNs. The proposed MCSOA leverages chaotic maps and seagull optimization algorithm principles to enhance global search capabilities and prevent local optima, offering a robust and efficient solution for routing optimization.

## III. MCSOA SOLVES THE TARFET ALLOCATION PROBLEM IN PSMSNS

In this section, we introduce the MCSOA to solve the task allocation problem in PSMSNs. The MCSOA leverages the advantages of chaos theory and quantum computing to enhance the optimization performance of the traditional SOA.

The Multi-Objective Chaotic Seagull Optimization Algorithm (MCSOA) is an advanced optimization technique inspired by the natural behaviors of seagulls, particularly their migration and attacking patterns. This algorithm integrates chaotic maps and multi-objective optimization principles to enhance the search capabilities and convergence properties of the traditional Seagull Optimization Algorithm (SOA). The MCSOA is designed to solve complex optimization problems by maintaining a balance between exploration and exploitation, ensuring a diverse and convergent set of solutions.

Seagulls exhibit distinct behaviors during their migration and hunting processes. These behaviors can be mathematically modeled to create effective optimization strategies:

Migration: Seagulls migrate in groups towards regions with abundant resources. This behavior is modeled to represent the exploration phase of the algorithm.

Attacking: During hunting, seagulls attack their prey using spiral flight patterns. This behavior is used to simulate the exploitation phase, focusing on refining solutions to achieve optimal results.

### *A. Algorithm Initialization*

A population of seagulls is initialized with random positions within the search space, where each seagull represents a potential solution to the optimization problem. These positions are distributed to cover the entire search space, ensuring diversity.

$$x_i = x_{min} + (x_{max} - x_{min}) \cdot rand(0,1) \tag{12}$$

In (12), where:

$x_i$ is the position of the $i$-th seagull,

$rand(0,1)$ generates a random number between 0 and 1.

### *B. Migration Phase*

During the migration phase, seagulls adjust their positions based on the best-known positions in the group. This phase ensures global exploration of the search space:

$$x_i(t+1) = x_i(t) + \alpha \cdot \text{rand}(0,1) \cdot (x_{\text{best}} - x_i(t)) \tag{13}$$

In (13), where:

$x_i(t)$ is the current position,

$x_{best}$ is the position of the best-performing seagull,

$\alpha$ is a scaling factor.

### *C. Attacking Phase*

In the attacking phase, seagulls perform a spiral flight pattern to refine their search around promising regions. This phase focuses on local exploitation:

$$x_i(t+1) = x_i(t) + \beta \cdot \text{rand}(0,1) \cdot (x_{\text{best}} - x_i(t)) + \gamma \cdot \sin(2\pi \cdot t) \tag{14}$$

In (14), where:

$\beta$ and $\gamma$ are constants that control the intensity of the search,

$\sin(2\pi \cdot t)$ introduces a chaotic component to avoid local optima.

### *D. Chaotic Map Integration*

To enhance the exploration capabilities, MCSOA incorporates chaotic maps, which introduce non-linear dynamics into the search process. Commonly used chaotic maps include the Logistic map and Tent map:

$$x_{n+1} = r \cdot x_n \cdot (1 - x_n) \tag{15}$$

In (15), where:

$r$ is the control parameter,

$x_n$ is the chaotic variable at iteration $n$.

### *E. Multi-Objective Optimization*

MCSOA uses a roulette wheel selection method to select efficient archived solutions, mimicking the migration and attacking patterns of seagulls.The fitness of each solution is evaluated based on multiple objectives, ensuring a balance between different performance metrics.

### *F. Algorithm Steps*

The MCSOA follows the steps outlined below:

- Initialization: Initialize the population of seagulls. Set parameters for migration, attacking, and chaotic maps.

- Evaluation: Evaluate the fitness of each seagull based on multiple objectives.
- Migration Phase: Update positions using the migration model. Incorporate chaotic maps for enhanced exploration.
- Attacking Phase: Refine positions using the attacking model. Ensure diversity and convergence.
- Update Archive: Store non-dominated solutions in the dynamic archive. Select effective solutions using roulette wheel selection.
- Termination: Repeat steps 2-5 until the stopping criterion is met, such as reaching a maximum number of iterations or a convergence threshold.

## IV. Simulation And Resault Analysis

This section outlines the simulation setup and results used to assess MCSOA's effectiveness in optimizing routing for PSWSNs. We compared MCSOA with traditional GA and PSO methods using simulations on a computer equipped with an Intel(R) Core(TM) i7-10700K CPU @ 3.80GHz.

For our experiments, we considered various scenarios with different numbers of nodes (50, 100, 150, and 200). Each algorithm was executed for 300 iterations in each scenario. Key metrics evaluated included node lifespan, energy consumption, and data packets received by the base station (BS).

### *A. Node Lifespan*

The lifespan of nodes is a crucial metric in WSNs, as it directly impacts network reliability and maintenance costs. The results for node lifespan across different numbers of nodes are summarized in Fig.1.

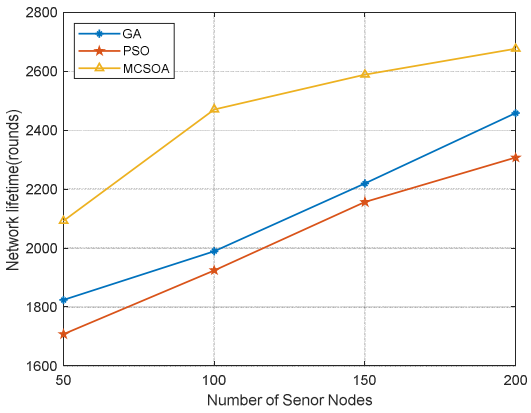



Fig. 1. Simulation Results of Node Lifetime: 50, 100, 150, and 200 nodes

From Fig.1, it is evident that the MCSOA consistently outperforms both GA and PSO across all node configurations. For instance, with 50 nodes, the MCSOA extends the node lifespan by 15.6% compared to GA and by 22.6% compared to PSO. This trend continues with larger node counts, demonstrating the effectiveness of MCSOA in enhancing network longevity.

### *B. Energy Consumption*

Efficient energy usage is essential for the sustainability of WSNs. Fig.2 presents the average energy consumption per node for each algorithm.

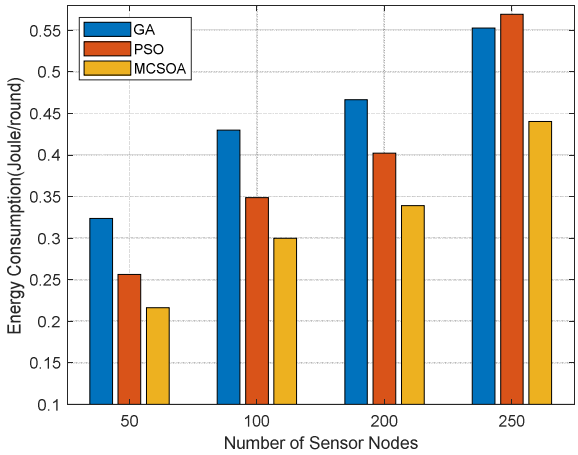



Fig. 2. Simulation Results for Energy Consumption: 50, 100, 150, and 200 nodes

As seen in Fig.2, MCSOA demonstrates superior energy efficiency, consuming significantly less energy than both GA and PSO. For example, with 50 nodes, MCSOA reduces energy consumption by 33.3% compared to GA and by 16.1% compared to PSO. This reduction in energy usage translates to prolonged operational periods for the network.

### *C. Number of Data Packets Received by the BS*

The ability of a WSN to transmit data reliably to the BS is critical for its functionality. Fig.3 displays the number of data packets received by the BS for each algorithm.

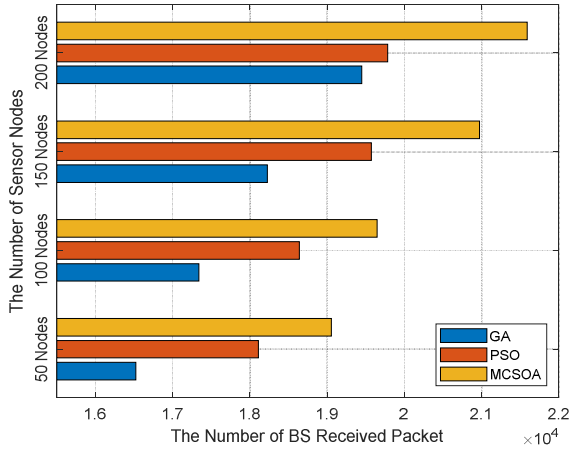



Fig. 3. Simulation Results for Number of Data Packets Received by the BS: 50, 100, 150, and 200 nodes

Fig.3 illustrates that MCSOA enables the transmission of more data packets to the BS compared to GA and PSO. For instance, with 50 nodes, MCSOA improves data packet reception by 15.3% over GA and by 5.2% over PSO. This improvement ensures more reliable and efficient data communication within the network.

The simulation results highlight MCSOA's enhanced performance compared to conventional GA and PSO algorithms. MCSOA not only extends the node lifespan but also optimizes energy consumption and enhances data packet transmission. The incorporation of chaotic mapping and seagull behavior in MCSOA significantly contributes to its ability to explore and exploit the solution space effectively, leading to higher performance metrics across various scenarios.

These findings validate the effectiveness of MCSOA in optimizing routing in WSNs for power systems, making it a promising approach for enhancing the reliability and efficiency of such networks. Future work could explore further optimizations and adaptations of MCSOA to different network configurations and environmental conditions.

## V. CONCLUSIONS

In this paper, we proposed the MCSOA for optimizing routing in high-reliability PSWSNs context. The MCSOA integrates chaotic maps with seagull optimization, enhancing the algorithm's capability to prevent getting trapped in local optima and strike a balance between exploration and exploitation.

Through extensive simulations and comparative analysis, MCSOA demonstrated superior performance in terms of convergence speed, solution diversity, and reliability when compared to existing algorithms. The chaotic maps incorporated into MCSOA significantly improved the search efficiency and robustness, making it a promising solution for the complex nature of WSNs in power systems.

Moreover, the dynamic archive mechanism employed by MCSOA ensures the retention of Pareto optimal solutions, further contributing to the algorithm's effectiveness in multi-objective optimization scenarios. The empirical results validated the algorithm's applicability and efficiency in real-world PIoT environments.

In summary, MCSOA provides a novel and effective approach to routing optimization in PIoT, offering a high-reliability solution that meets the stringent requirements of modern power systems. Future work will focus on further refining the algorithm, exploring additional chaotic maps, and expanding its application to other optimization problems in the field of cybersecurity and beyond.

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